

Evaluating and Fine-Tuning Large Language Models for Question Answering on DNA Methylation

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Abstract

We study how well modern large language models (LLMs) answer questions about DNA methylation, and whether a small open-source model like Qwen3-1.7B can be fine-tuned to match a frontier model on this specialized domain. We construct a 400-question multiple-choice benchmark from the primary literature on DNA methylation and epigenetic aging. We evaluate eight model configurations: base (non-thinking) and thinking of Qwen3-1.7B and Qwen3-8B, three fine-tuned variants of Qwen3-1.7B, and Gemini 2.5 Flash as a frontier reference. The fine-tuning recipes include: (1) QA-only, (2) combined QA+summaries (single-stage), and (3) a two-stage fine-tune that first trains on the combined QA+summaries mixture and then realigns with QA-only training. Overall, Gemini 2.5 Flash leads at 78.2%. The two-stage Qwen3-1.7B reaches 74.3%, outperforming Qwen3-8B in thinking mode despite being non-thinking and roughly one-fifth its size. Across both Qwen3 models, thinking mode yields 3.3–7.0 point improvements, and single-stage QA improves the base model by 11.5 points, whereas the single-stage QA+summaries *hurts* accuracy by 7.8 points from the base. These results suggest that two-stage fine-tuning can be more effective than model scale or reasoning mode for adapting small open models to specialized biomedical QA.

1 Introduction

1.1 Background and motivation

DNA methylation is one of the most actively studied topics in modern biology. Concretely, it refers to chemical marks that cells attach to specific sites along the DNA sequence (known as CpG sites). These marks sit on top of the DNA without changing the underlying code and act as switches to control whether nearby genes are on or off. The pattern of these marks across the genome shifts as people age, looks systematically different in cancerous versus healthy tissue, and carries information about disease risks and likely responses to treatments. The field received enormous momentum after the publication of the first genome-wide human methylomes around 2009 [1], and the subsequent development of “epigenetic clocks” that estimate biological age from methylation patterns [2, 3]. It has also been used to track disease risk, and evaluate the effects of lifestyle and therapeutic interventions [5, 6].

Meanwhile, LLMs are increasingly proposed as medical assistants [7, 8]. Methylation researchers and professionals need agents to help read, synthesize, and reason over methylation literature and

patient-level methylation data. Yet, despite the enormous promise in DNA methylation, it is still a young field: methylation-specific research constitutes a small fraction of the total biomedical literature, and an even smaller fraction of the web-scale text on which modern large language models (LLMs) are trained. The gap between what modern LLMs have been exposed to and what methylation researchers need them to do is what this project examines.

Two observations set up the rest of the paper. First, a model’s score on a general biomedical benchmark (e.g., MedQA, PubMedQA, MMLU-medicine) tells a researcher little about whether that model can be trusted on the specialized methylation literature they actually read; those benchmarks are dominated by general clinical content and rarely probe molecular-biology subfields. A methylation-specific benchmark is therefore needed. Second, if a small open-weights model can be fine-tuned to close the gap to a frontier model on this narrow domain, the practical implications are substantial: locally hosted models preserve patient-data privacy, are cheaper to run at scale, and are more reproducible than commercial APIs whose weights change over time.

1.2 Problem statement

This project addresses two coupled questions:

1. **How well do current LLMs answer questions about DNA methylation?** Specifically, we compare a frontier model (Gemini 2.5 Flash) to two sizes of a capable open-source family (Qwen3-1.7B and Qwen3-8B), across four cognitive question types.
2. **Can a small open model be fine-tuned to match a frontier model on this domain, and if so, how?** We compare three fine-tuning recipes: (1) QA-only, (2) combined QA+summaries, and (3) a two-stage fine-tune (QA+summaries, then QA alone), to isolate which kind of domain adaptation actually helps.

1.3 Data and approach

The benchmark comprises 400 multiple-choice questions generated from a curated set of peer-reviewed papers on DNA methylation and epigenetic aging, with each question manually labeled by cognitive category. Fine-tuning data is drawn from a separate held-out set of methylation-related literature (both as raw summaries and as generated QA pairs). Evaluation is done by zero-shot multiple-choice inference, with stratified accuracy as the primary metric and cross-model overlap statistics as secondary metrics.

1.4 Road map of the report

Section 2 describes the construction of the benchmark and fine-tuning datasets, with descriptive statistics and a discussion of why we pivoted away from an earlier tabular-memorization approach. Section 3 describes the eight evaluated model configurations, the fine-tuning procedures, and the inference setup. Section 4 presents results: overall accuracy, stratified accuracy by cognitive type, and question-level overlap between models. Section 5 discusses what these results say about the roles of scale, reasoning mode, and fine-tuning in specialized biomedical QA. Section 6 concludes with recommendations and suggestions for future work.

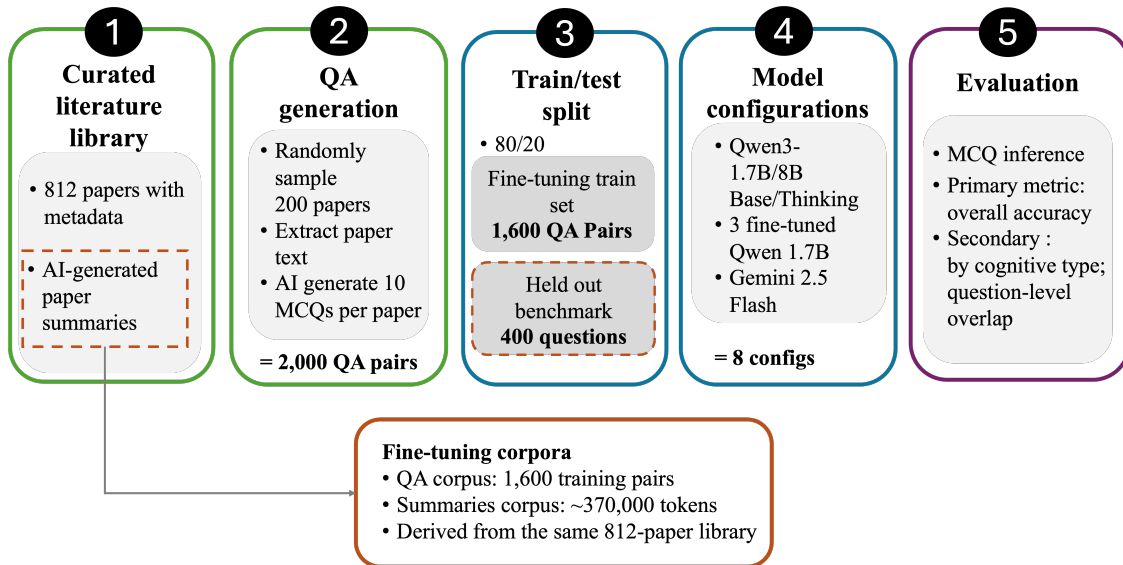


Figure 1: End-to-end workflow for constructing and evaluating the methylation QA benchmark. A curated 812-paper methylation library is used to generate 2,000 multiple-choice QA pairs, which are split into 1,600 fine-tuning examples and a 400-question held-out benchmark. The same benchmark is then used to evaluate eight model configurations across overall accuracy, cognitive-category accuracy, and question-level overlap.

2 Benchmark

2.1 Benchmark construction

Figure 1 summarizes the full experimental workflow. The project begins with a curated methylation literature library, converts a random subset of papers into multiple-choice QA pairs, splits those pairs into fine-tuning and held-out evaluation sets, and then compares eight model configurations on the same 400-question benchmark.

The benchmark and fine-tuning data are both derived from a single curated library of 812 peer-reviewed PDFs on DNA methylation and epigenetic aging, compiled by the Higgins-Chen Lab at Yale. The library is organized into roughly 20 thematic subdirectories (e.g., clock–trait associations, mechanism, risk, non-blood tissues, statistical methods, preprocessing tools) and tracked by a master metadata table recording DOI, PMID, title, year, authors, journal, and an AI-generated summary for each paper, also produced by the Higgins-Chen Lab.

From this library, 200 papers were selected at random for multiple-choice question generation. For each selected paper, text was extracted with PyMuPDF (truncated to 800,000 characters) and passed to Gemini 2.5 Flash with a prompt requesting 10 four-choice MCQs tagged with `paper_source`, topic category, and cognitive type (Factual Recall, Conceptual Understanding, Methodological Reasoning, or Quantitative/Analytical). Generated JSON was parsed and validated for required fields and answer normalization, yielding 2,000 QA pairs. These curated QA pairs were then partitioned 80/20 by a random split (seed 1234): 1,600 pairs form the fine-tuning train set and the remaining 400 form the held-out *benchmark* test set used throughout this paper. The split is at the QA level rather than the paper level, so a given paper may contribute questions to both partitions — though no individual question appears in both.

Each question was tagged with one of four cognitive categories:

- **Factual Recall (162 Q, 40.5%)**. Specific numeric results, dataset names, CpG site identifiers, or published figures that the model must recall from the literature.
- **Conceptual Understanding (116 Q, 29.0%)**. Mechanistic or biological explanations, e.g., why promoter methylation silences transcription.
- **Methodological Reasoning (70 Q, 17.5%)**. Experimental design, statistical, and analytical choices; e.g., appropriate controls for a methylation array study.
- **Quantitative / Analytical (52 Q, 13.0%)**. Numerical reasoning over results, e.g., interpreting a regression coefficient or computing an age delta.

Figure 2 visualizes the composition of the benchmark.

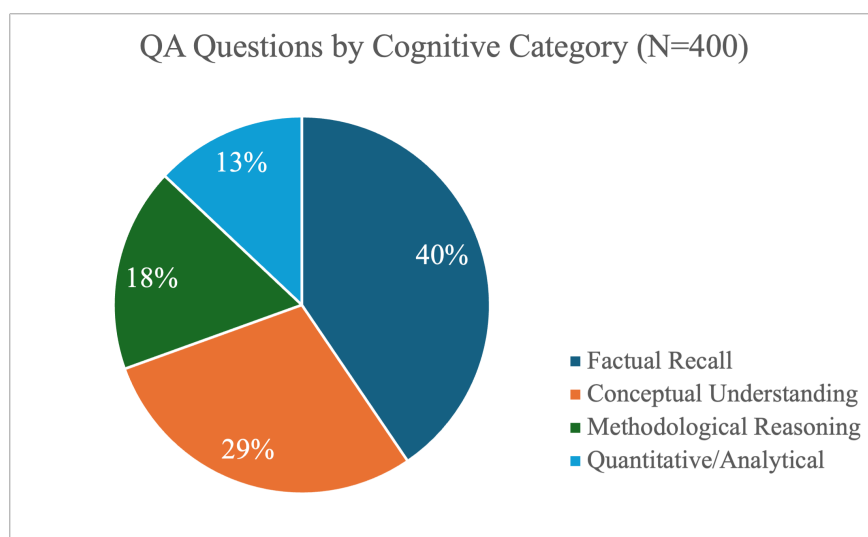


Figure 2: Distribution of the 400 benchmark questions across four cognitive categories. Factual recall dominates, reflecting the empirical nature of the methylation literature.

A representative question from the *Conceptual Understanding* category:

Literature-Based QA — Conceptual Understanding

Q: Why does DNA methylation at CpG islands in gene promoter regions typically lead to transcriptional silencing?

A: Methylation of promoter CpG islands recruits methyl-binding domain (MBD) proteins and chromatin remodelers that promote a closed, heterochromatic state. This blocks transcription factor binding and RNA polymerase II recruitment, silencing the associated gene.

This question rewards a model that has internalized the general mechanism (MBD recruitment → chromatin closure → silencing) rather than one that has memorized any specific study. It is the category of question we most want an LLM assistant to answer well.

2.2 From tabular memorization to literature-based QA

An early iteration of this project generated QA pairs directly from tabular methylation datasets — spreadsheets associating CpG sites with phenotypes such as insulin resistance, smoking status, or age. The goal was to test whether a small language model could be fine-tuned to recall specific CpG identifiers from these tables, and to check its answers against ground truth. We fine-tuned Qwen3-1.7B on the tabular QA pairs and evaluated the resulting model on held-out queries drawn from the same tables. In practice this experiment was uninformative. A representative failure from the fine-tuned model looked like this:

Tabular Memorization QA — fine-tuned model output

Prompt: List CpG sites linked to insulin resistance.

Ground truth: The 4 CpG sites associated with insulin resistance are cg00004939, cg00007800, cg00008819, cg00017601.

Fine-tuned model prediction: The CpG site associated with insulin resistance is cg00003228. ✗ Hallucinated identifier

Even after fine-tuning directly on the target tables, the model hallucinated a plausible-looking CpG identifier that is absent from the ground-truth set. This pattern held across phenotypes: the fine-tuned model reliably produced output in the correct format (a **cg**-prefixed 8-digit identifier) but failed to reproduce the specific identifiers it had been trained on. We interpret this as evidence that memorizing a large table of arbitrary alphanumeric identifiers is a poor fit for small-model fine-tuning: there is no compressible structure to learn and no semantic cue linking a phenotype to its CpG set, so the model instead learns the surface format and generates identifiers that look plausible but are wrong.

More fundamentally, this is the wrong question to ask of a language model. Tabular memorization tests whether a model can serve as a lookup table for a specific spreadsheet — a task at which conventional databases are faster, cheaper, and exactly correct. We therefore pivoted to a literature-based benchmark in which questions test reasoning over scientific content that is at least plausibly inferable from training exposure or from the question text itself. This is the form of question where a language model can actually add value that a database cannot.

2.3 Why multiple-choice?

A natural alternative to multiple-choice evaluation is free-response question answering, in which the model generates an open-ended answer that is then graded for correctness. We considered this format and ultimately rejected it for three reasons.

First, **evaluating free-response answers is tedious and often subjective**. An answer may sound plausible but be factually wrong, be partially correct in ways that require scientific judgment to score, or be viewed by one grader as solid and another as incomplete. Manually grading 400 questions $\times$ 8 models is prohibitively expensive and introduces inter-rater disagreement. Alternatively, using another LLM grader makes it less labor-intensive but imports another layer of biases, or worse, creates a circularity when the judge is in the same family as the model under test. Multiple-choice avoids all of this, as each question has one correct letter, so

Second, **free-response scoring rewards fluency over accuracy**. Models that produce confident, well-organized, vaguely-correct-sounding hallucinations tend to be scored more favorably

than models that produce terse but precise answers, even when the latter are factually better. This is a documented failure mode of LLM-as-judge evaluations and is the exact failure we want the benchmark to detect.

Third, **multiple-choice is unambiguous and reproducible**. Every question has exactly one correct answer, so scoring is just string matching and results are easy to reproduce, which is especially important for a benchmark that we hope will be used to track progress over time.

The cost of this choice is that multiple-choice scoring does not capture several properties of real-world use: the quality of free-form explanations, consistency under paraphrase, and calibration of expressed confidence. We view these as important targets for follow-up evaluation (see Section 7). For now, as our primary benchmark, multiple-choice offers objectivity, reproducibility, and ease of cross-model comparison, making it exactly what a *first* benchmark for an under-studied scientific domain needs.

3 Experimental Setup

3.1 Models

Table 1 lists the eight configurations evaluated. Qwen3-1.7B was chosen because it is small enough to fine-tune on a single consumer-grade GPU, making it representative of models accessible to individual research labs. Qwen3-8B is not fine-tuned in this project; it serves as a scale-only reference within the same model family. Gemini 2.5 Flash is the frontier reference, accessed via API.

#	Model	Mode	Fine-tune recipe
1	Qwen3-1.7B	Base	-
2	Qwen3-1.7B	Thinking	-
3	Qwen3-1.7B	Base	QA-only
4	Qwen3-1.7B	Base	QA+Summaries
5	Qwen3-1.7B	Base	Two-stage
6	Qwen3-8B	Base	-
7	Qwen3-8B	Thinking	-
8	Gemini 2.5 Flash	Base	-

Table 1: The eight model configurations evaluated in this study. Fine-tuning recipes are described in Section 3.3.

3.2 Fine-tuning corpus

The fine-tuning data comes from the same 812-paper library described in Section 2.1. The library’s AI-generated summaries form the **summaries corpus**, totaling approximately 370,000 tokens. The **QA corpus** is the 1,600-pair training split produced by the same 80/20 partition that yields the benchmark test set (Section 2.1).

The two corpora are kept separate by design: summaries provide broad domain prose, while QA pairs provide task-format examples. How they are combined, and in what order, is the central variable across the three fine-tuning recipes.

3.3 Fine-tuning procedure

All fine-tuning uses the AdamW optimizer with a cosine learning rate schedule and `warmup_ratio=0.05`. We chose to have `loss_on_response_only` (loss computed on answer tokens only, not on the prompt) for two reasons. First, Qwen3-1.7B already has strong instruction-following capabilities from pre-training, so the prompt tokens carry no additional learning signal for our purposes. Second, since the QA responses are single letters, computing loss over the hundreds of tokens in the question prompt would introduce noise that dilutes the gradient signal we actually care about.

The three recipes differ in training data, epoch count, and effective batch size:

QA-only. Trained on the 1,600-pair QA training split for 120 epochs. Peak learning rate 1×10^{-5} , effective batch size 64 (`per_device_batch_size=8`, `gradient_accumulation_steps=8`).

QA+Summaries, single-stage. Trained on the union of the 1,600 QA pairs and 812 paper summaries for 100 epochs. Peak learning rate 1×10^{-5} , effective batch size 64 (`per_device_batch_size=2`, `gradient_accumulation_steps=32`).

Two-stage. We reused the single-stage QA+Summaries recipe as the first stage. The stage 1 checkpoint at step 3100 is used as the initialization for a second fine-tuning run using the same configuration as the QA-only recipe.

The motivation for the two-stage design is separation of concerns: stage 1 acquires domain knowledge across two formats (prose and QA), while stage 2 re-aligns the model to the multiple-choice output format. Results in Section 4 show that this separation is necessary — combining the same data in a single stage performs substantially worse.

3.4 Evaluation metrics

Our primary metric is multiple-choice accuracy over all 400 questions. Secondary analyses include:

- **Stratified accuracy** by cognitive category.
- **Question-level overlap:** the proportion of questions answered correctly by all models (“easy”), by none (“hard”), and uniquely by a single model.
- **Pairwise delta** between configurations that differ by a single intervention (thinking on/off; fine-tuning on/off; 1.7B vs 8B), used to isolate the effect of each intervention.

3.5 Inference setup

All eight configurations are evaluated on the same held-out test split. Every prompt frames the model as a domain expert (“You are an expert in DNA methylation”), presents the question stem followed by four labelled answer choices (A–D), and instructs the model to produce a single uppercase letter. Responses with no valid letter are scored as incorrect.

Base models and all fine-tuned variants use greedy decoding (`do_sample=False`, `temperature=0.0`, `max_new_tokens=16`). Thinking variants use stochastic sampling (`temperature=0.6`, `top_p=1.0`, `max_new_tokens=4096`) with `thinking_budget=512`, following Qwen3’s recommended inference settings, and the prompt is reformulated to elicit a brief 2–3 sentence reasoning trace followed by a final answer in the format `ANSWER: X`. Gemini 2.5 Flash is queried with `temperature=0.0`,

`thinking_budget=0` (extended thinking disabled), and `max_output_tokens=1024`. Random seeds are 1234 for all models. The full prompt template is given in Appendix ??.

4 Results & Analysis

4.1 Overall accuracy

Table 2 and Figure 3 report overall accuracy for each of the eight configurations. Gemini 2.5 Flash leads at 78.2%. The two-stage fine-tuned Qwen3-1.7B is second at 74.3%, ahead of Qwen3-8B in thinking mode (71.5%) despite being roughly one-fifth its parameter count and the latter being evaluated in thinking mode. The main empirical result is therefore not simply that larger or frontier models perform better. More importantly, we see that a carefully fine-tuned 1.7B model can exceed a much larger model from the same family. Therefore, choosing the right fine-tuning strategy can substantially change model performance.

Because the benchmark is not evenly distributed across cognitive categories, we also computed a category-normalized accuracy by averaging the four category accuracies with equal weight. This normalization does not change the model ranking: Gemini remains first at 76.6%, followed by the two-stage Qwen3-1.7B model at 73.1% and Qwen3-8B Thinking at 72.9%. However, the gap between the best open models and Gemini narrows slightly, suggesting that Gemini’s overall lead is partly driven by its strength on factual recall, the largest category in the benchmark.

Model	Correct / 400	Overall accuracy	Category-normalized accuracy	Rank
Gemini 2.5 Flash	312	78.2%	76.6%	1
Qwen3-1.7B — Two-stage fine-tune	297	74.3%	73.1%	2
Qwen3-8B — Thinking	286	71.5%	72.9%	3
Qwen3-8B — Base	273	68.3%	68.3%	4
Qwen3-1.7B — QA-only	269	67.3%	66.7%	5
Qwen3-1.7B — Thinking	251	62.8%	64.1%	6
Qwen3-1.7B — Base	223	55.8%	57.6%	7
Qwen3-1.7B — QA+Summaries	192	48.0%	48.0%	8

Table 2: Overall accuracy and category-normalized accuracy across all eight model configurations, sorted by rank.

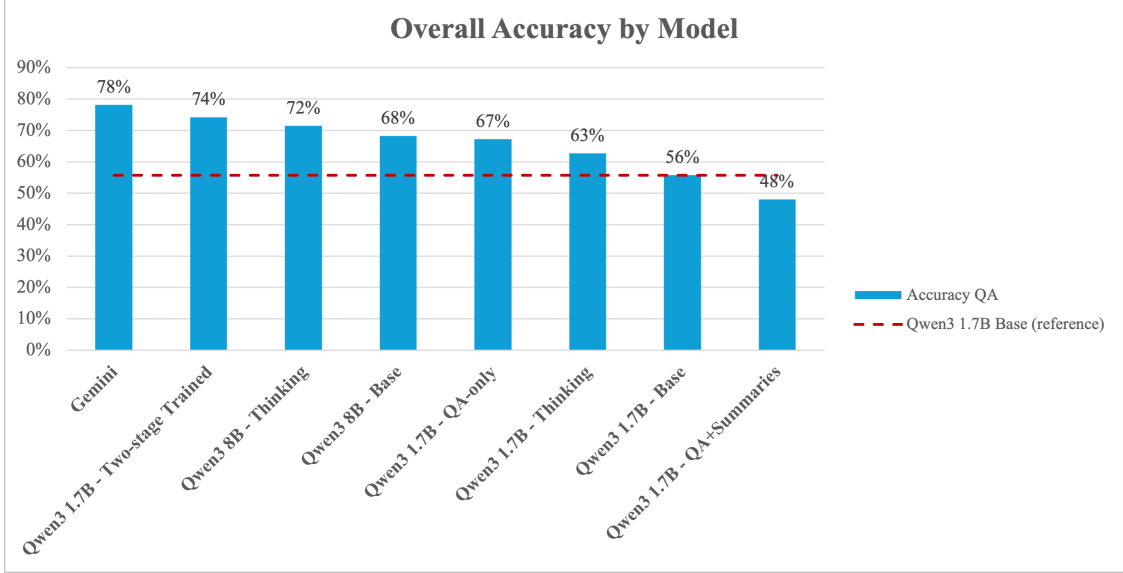


Figure 3: Overall benchmark accuracy across all eight evaluated model configurations. Gemini 2.5 Flash performs best overall, while the two-stage Qwen3-1.7B fine-tune is the strongest open-source configuration and outperforms Qwen3-8B Thinking despite being much smaller.

4.2 Accuracy by cognitive category

Table 3 compares accuracy by the four cognitive categories. This is visualized in the heatmap in Figure 4. Model differences are largest on factual recall and quantitative questions, whereas strong models cluster closely on conceptual and methodological questions. Gemini leads on factual recall and conceptual understanding, while Gemini, Qwen3-8B Thinking, and the two-stage Qwen3-1.7B model tie on methodological reasoning. Qwen3-8B Thinking performs best on quantitative/analytical questions.

Model	Factual	Conceptual	Methodological	Quantitative	Overall
Gemini 2.5 Flash	74.5%	88.8%	81.4%	61.5%	78.2%
Qwen3-1.7B — Two-stage fine-tune	67.9%	87.1%	81.4%	55.8%	74.3%
Qwen3-8B — Thinking	59.3%	85.3%	81.4%	65.4%	71.5%
Qwen3-8B — Base	60.5%	80.2%	72.9%	59.6%	68.3%
Qwen3-1.7B — QA-only	60.5%	77.6%	78.6%	50.0%	67.3%
Qwen3-1.7B — Thinking	48.1%	79.3%	77.1%	51.9%	62.8%
Qwen3-1.7B — Base	43.2%	67.2%	70.0%	50.0%	55.8%
Qwen3-1.7B — QA+Summaries	37.7%	62.1%	61.4%	30.8%	48.0%
Mean across models	56.5%	78.4%	75.5%	53.1%	65.5%

Table 3: Accuracy by cognitive category across all eight evaluated model configurations. Bold entries indicate the best model in each category.

Several patterns stand out. First, Gemini has the biggest advantage in factual recall questions, achieving 74% accuracy, compared with 67.9% for the two-stage Qwen3-1.7B model and 60.5% for Qwen3-8B. A lot of these factual recall questions ask for specific findings, database details, cohort characteristics, implementation details, or published numerical results that are difficult for a small

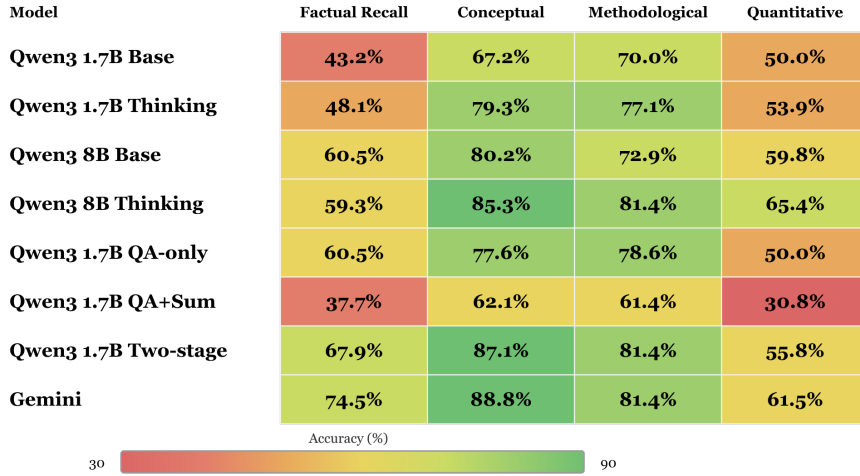


Figure 4: Accuracy by cognitive category for each model configuration. Warmer cells indicate lower accuracy and greener cells indicate higher accuracy. The heatmap shows that conceptual and methodological questions are comparatively easier across models, while factual-recall and quantitative/analytical questions remain harder. The two-stage Qwen3-1.7B model is especially competitive on conceptual and methodological reasoning, while Qwen3-8B Thinking performs best on quantitative questions.

model to infer from reasoning alone. The results, however, suggest that fine-tuning successfully transfers some domain-specific knowledge into the small model. Factual recall is also the largest category in the benchmark, so Gemini’s strength on factual recall contributes substantially to its overall lead.

Second, the best open-source models are much more competitive on conceptual and methodological questions. On conceptual understanding, the two-stage Qwen3-1.7B model reaches 87.1%, only 1.7 points below Gemini. On methodological reasoning, the two-stage Qwen3-1.7B model, Qwen3-8B Thinking, and Gemini all tie at 81.4%. This seems to suggest that the fine-tuned small model is not merely memorizing answer patterns; it is also acquiring useful domain concepts and methodological structure.

Third, quantitative questions are the hardest for most models. Qwen3-8B Thinking performs best at 65.4%, followed by Gemini at 61.5%. The two-stage Qwen3-1.7B model reaches only 55.8%, despite its strong overall performance. This is also the only category in which Gemini is not the top model, suggesting that thinking-mode inference may be particularly useful for questions involving calculation or multi-step reasoning.

4.3 Question-level overlap

Table 4 reports overlap rates within each cognitive category. Figure 5 complements it by separating two kinds of question-level behavior. Panel A shows shared difficulty: conceptual and methodological questions are often answered correctly by all models, whereas factual-recall and quantitative questions contain more items that no model solves. Panel B shows model-specific strengths: Gemini has the clearest unique advantage on factual recall, but the two-stage Qwen3-1.7B model and Qwen3-8B Base also answer a small number of questions that all other models miss.

Category	All correct	None correct	Only Gemini	Only 1.7B Two-Stage	Only 8B Base	Only 1.7B QA	Only 1.7B Base	Only 1.7B Sum	Only 8B Think	Only 1.7B Think
Conceptual	48.3%	0.9%	0.0%	1.7%	0.0%	1.7%	0.0%	0.0%	0.9%	0.0%
Methodological	45.7%	4.3%	0.0%	0.0%	0.0%	0.0%	1.4%	1.4%	0.0%	0.0%
Quantitative	23.1%	7.7%	1.9%	1.9%	1.9%	0.0%	0.0%	0.0%	1.9%	0.0%
Factual Recall	21.6%	6.2%	4.3%	2.5%	2.5%	1.2%	0.6%	0.6%	0.0%	0.6%
All Questions	33.8%	4.5%	2.0%	1.8%	1.3%	1.0%	0.5%	0.5%	0.5%	0.3%

Table 4: Question-level overlap by cognitive category. Values are percentages within each cognitive category; for example, 48.3% (56/116) of conceptual questions were answered correctly by all models. The “All questions” row reports percentages over all 400 benchmark questions. “Only” means that exactly one model answered the question correctly.

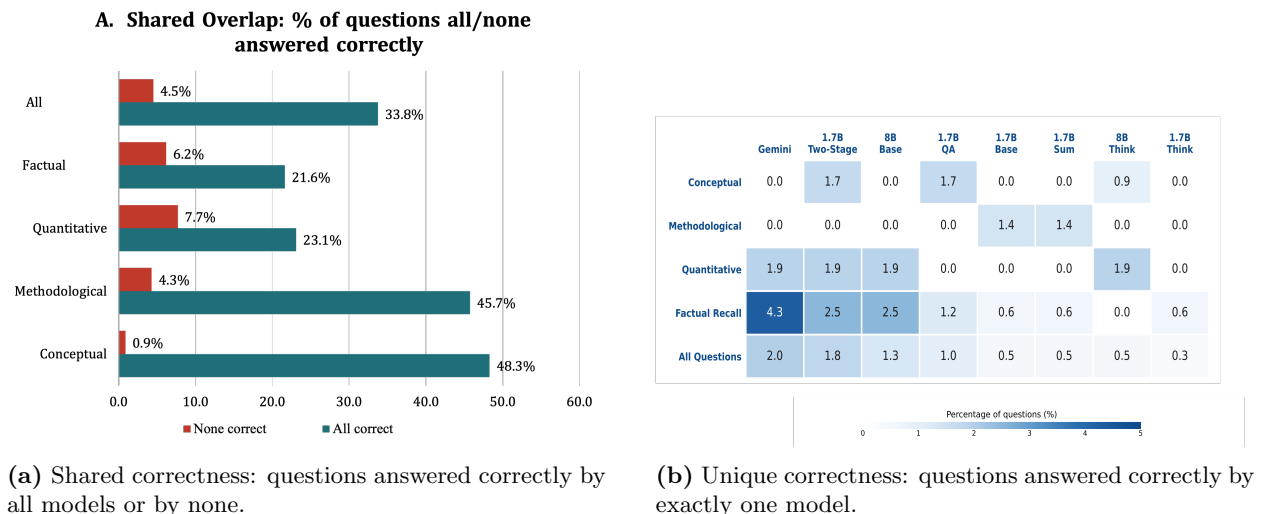


Figure 5: Question-level overlap patterns by cognitive category. Panel A shows that conceptual and methodological questions have the highest all-model-correct rates, while factual-recall and quantitative questions have more questions that no model answers correctly. Panel B shows that Gemini’s unique correct answers are concentrated in factual recall, but other models also have small pockets of non-overlapping strengths.

Conceptual and methodological questions have the highest all-model-correct rates, at 48.3% and 45.7%, respectively, which also mirrors their highest overall accuracy across models revealed in Table 3. By contrast, only 21.6% of factual-recall questions and 23.1% of quantitative questions are answered correctly by all models. Similarly, the no-model-correct rate is also highest for quantitative questions (7.7%) and factual-recall questions (6.2%). Gemini’s unique correct answers are concentrated in factual recall: 4.3% of factual-recall questions are answered correctly only by Gemini, compared with 0.0% of conceptual and methodological questions. This helps explain why Gemini has the highest overall accuracy, given that factual recall is the largest category in the benchmark.

At the same time, unique correct answers are not exclusive to Gemini. The two-stage Qwen3-1.7B model has unique wins in conceptual, quantitative, and factual-recall questions, while Qwen3-8B Base has unique wins in quantitative and factual-recall questions. It is possible that different model

configurations retain some non-overlapping strengths, even when their overall accuracies differ.

4.4 Effect-size decomposition

The results above separate three possible sources of improvement: model scale, thinking-mode inference, and fine-tuning. Scaling from Qwen3-1.7B Base to Qwen3-8B Base improves accuracy from 55.8% to 68.3%, a gain of 12.5 percentage points. Enabling thinking mode produces smaller but consistent gains: Qwen3-1.7B improves from 55.8% to 62.8% (+7.0 points), while Qwen3-8B improves from 68.3% to 71.5% (+3.3 points).

Fine-tuning produces a much larger spread than either scale or thinking. Three results stand out:

First, **Single-stage QA+Summaries fails**. Training on a mixture of QA pairs and free-form summaries in a single pass *decreased* accuracy from 55.8% to 48.0% (-7.8 percentage points). We interpret this as a form of task-format interference: mixing free-form prose and multiple-choice QA in the same optimization confuses the model’s output distribution. This is amplified by a large imbalance in token counts: the 812 paper summaries contribute substantially more tokens than the 1,600 QA pairs, so free-form prose dominates the training signal and the small model learns much less from the single-letter answers.

Second, **QA-only fine-tuning helps moderately**. Fine-tuning directly on QA pairs lifted the 1.7B model from 55.8% to 67.3% (+11.5 percentage points), enough to essentially match the base 8B model (68.3%). The effect is modest but consistent across categories.

Two-stage fine-tuning wins. The two-stage recipe reached 74.3%, a gain of 18.5 percentage points over base and 7.0 over QA-only. The same data that decreased performance when trained in a single pass *helps* substantially after the second stage task-aligned training is added. The interpretation is even though the first stage degrades task format, the model still gains in usable domain knowledge from the summary exposure. The second QA-only stage recovers the format while retaining the knowledge gained in stage one. Therefore, the two stages are complementary and the effects compound when trained in sequence. Section ?? examines the single-stage failure mode in more detail.

These comparisons suggest that, for this specialized biomedical QA benchmark, domain adaptation can matter as much as or more than raw parameter count. The strongest evidence is that the two-stage Qwen3-1.7B model outperforms Qwen3-8B Thinking despite being substantially smaller and not using thinking-mode. Of course, this does not mean scale and thinking are not important. Rather, it shows that the structure of fine-tuning is the largest driver of performance among the variants tested in this study.

The category-level results further clarify where these gains come from. Fine-tuning is especially effective on conceptual and methodological questions, where the model can learn recurring biological mechanisms, experimental designs, and analysis patterns. It is less sufficient for quantitative and factual-recall questions, where the answer may depend on exact numbers, cohort details, or facts from a specific paper. This distinction is important for interpreting both the strengths and limits of small-model specialization.

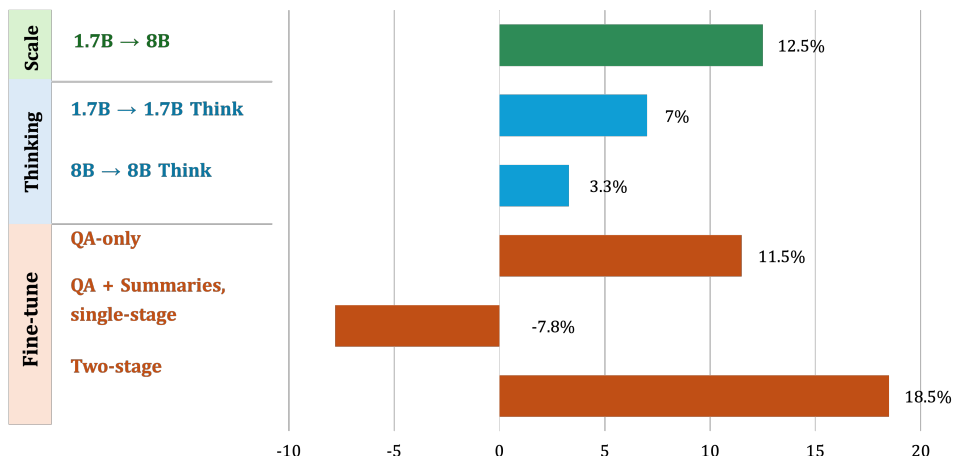


Figure 6: Decomposition of accuracy gains from scale, thinking-mode inference, and fine-tuning. Scaling from Qwen3-1.7B Base to Qwen3-8B Base improves accuracy substantially, and thinking mode gives smaller but consistent gains. The largest open-model improvement comes from two-stage fine-tuning, while single-stage QA+Summaries decreases performance, suggesting that domain exposure and task-format alignment must be separated.

5 Limitations and Future Work

This study has several limitations. First, the benchmark questions were generated from the literature using Gemini 2.5 Flash. Although this makes constructing a larger domain-specific benchmark efficiently, it may introduce bias toward the generator model. This is especially relevant because Gemini is also one of the evaluated models. Future work should include more expert-written or manually validated questions to reduce generator-model bias.

Second, the benchmark uses multiple-choice accuracy as the primary metric. This makes evaluation objective, reproducible, and easy to compare across models, but it does not capture all qualities needed in a real scientific assistant. In practice, researchers need models to explain their reasoning, cite evidence, express uncertainty, and remain robust to paraphrased or open-ended questions. Future evaluations should therefore include free-response QA, citation-grounded answers, calibration analysis, and robustness checks.

Some questions ask about exact cohort characteristics, numerical results, database details, preprocessing choices, or findings from a particular paper.

A third limitation is that some benchmark questions are paper-specific, while all models are evaluated in a closed-book setting. For example, the questions might ask exact numerical detail or a specific preprocessing step in a particular study. These questions are appropriate for testing whether a model can answer literature-grounded methylation questions, but in this project, they disadvantage our closed-book models. As a result, errors on factual-recall and quantitative questions should not be interpreted only as failures of domain understanding. In many cases, they reflect the difficulty of recalling source-specific details without retrieval. As a next step, the QA corpus should be reviewed to distinguish between general domain questions and paper-specific questions. This can produce more robust results. Alternatively, future studies should evaluate retrieval-augmented systems, where the model is given access to the relevant paper before answering.

Lastly, only Qwen3-1.7B is fine-tuned in this study. The results show that two-stage fine-tuning is highly effective for this small model, but it remains unclear whether the same pattern holds for larger open models or other model families. Future work could fine-tune larger Qwen models or compare open models from other families.

Overall, these limitations point to a clear next step: building a retrieval-augmented methylation QA system that combines a domain-adapted open model with grounded access to the source literature. Such a system would preserve the advantages of fine-tuning while addressing the above-mentioned limitations.

6 Conclusion

This project evaluates whether small open-source language models can be adapted for question answering in the specialized domain of DNA methylation. On a 400-question multiple-choice benchmark, Gemini 2.5 Flash achieves the highest overall accuracy at 78.2%, followed closely by a two-stage fine-tuned Qwen3-1.7B model at 74.3%, which outperforms Qwen3-8B Thinking despite being much smaller and using non-thinking inference.

Fine-tuning strategy matters. QA-only fine-tuning improves performance, single-stage QA+Summaries hurts performance, and two-stage fine-tuning produces the strongest open-model result. This suggests that summaries are useful to improve general domain familiarity, but needs to be followed by task-specific QA alignment.

Meanwhile, the benchmark reveals the limits of closed-book specialization. Models perform well on conceptual and methodological questions, but factual-recall and quantitative questions remain harder, especially when they require details or numerical values from specific studies. A more promising methylation QA assistant would therefore combine fine-tuning with retrieval: fine-tuning for domain fluency and answer formatting, and retrieval for the exact paper-level evidence needed for factual questions.

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Appendix

A. Prompt template

The multiple-choice prompt used at inference time:

You are an expert in DNA methylation. Answer the multiple-choice question below.

Only output a single uppercase letter: A, B, C, or D. Do not include any other text, punctuation, explanation, or markdown.

Question: {question}

A) {choice_A}

B) {choice_B}

C) {choice_C}

D) {choice_D}

Final answer:

For thinking-mode runs, the model is additionally permitted to emit a reasoning trace before the final letter; the answer parser takes the first letter token among {A, B, C, D} that appears after the final line of the trace, following "Final answer: ."